Ep 7 · Multi-Agent Design — Cheat Sheet

AI System Design · AI Engineering with Ajit · github.com/byruajit/ai-system-design

AN AGENT MUST EARN ITS TOKENS

Add an agent only for: **separation of duties** (tools/permissions) · **parallelism** · **independent check** (critic)

PATTERNS

Pattern	Use when
Single agent + tools	Default
Pipeline / workflow	Known process
Orchestrator → workers	Parallel sources
Generator ↔ critic	Quality/compliance gate
Free-form crew	Demos only

V2 RECIPE (CREDIT MEMO)

1. **Gather facts with code**
2. **Typed state**; facts read-only
3. One **drafter** + one **critic** (code checks + checklist)
4. **Retry cap** (2) → escalate
5. **Checkpoint** for human sign-off
6. Parallel branches only where slow

COST

Crew ≈ 5 agents × 4 turns × 6k = **120k tokens**
Graph ≈ **14k** → **~8.6× cheaper** *(assumptions)*

REPRODUCIBILITY

Fixed topology · temperature 0 · pinned models · structured output · checkpoints

DEBUG & AUDIT

Trace every node (inputs, outputs, model, tokens) · replay from checkpoints · prod failures → regression tests

SAY THIS

"Multi-agent is a pattern, not a goal. Here's what each agent buys us."